



December 2024

(For AD, AM, CB, CC, CS, IS)

Max. Marks: 100

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

20 Marks

- 1-

11. Which of this keyword must be used to inherit a class?
 - A) Super
 - B) this
 - C) extent
 - D) extends
12. What does the super keyword do?
 - A) It refers to the current object
 - B) It reference to the parent class
 - C) It calls a child class method
 - D) It is used to declare static variables
13. Which of these methods cannot be overridden?
 - A) Static methods
 - B) Public methods
 - C) Protected methods
 - D) Instance methods
14. What is the effect of using the final keyword with a method?
 - A) The method can be inherited but not overridden
 - B) The method cannot be inherited or overridden
 - C) The method can be inherited and overridden
 - D) The method must be abstract
15. Which method is used to close a JFrame when the close button is clicked?
 - A) setDefaultCloseOperation()
 - B) closeOnExit()
 - C) enableClose()
 - D) exitOnClose()
16. Which method is used to retrieve the detailed message of an exception?
 - A) printStackTrace()
 - B) getMessage()
 - C) toString()
 - D) getCause()
17. Which keyword is used to define an exception handling block?
 - A) catch
 - B) handle
 - C) try
 - D) throw
18. How can synchronization be achieved in Java threads?
 - A) Using the synchronized keyword
 - B) Using the volatile keyword
 - C) Using the final keyword.
 - D) Using the static keyword
19. What is the default priority of a thread in Java?
 - A) 1
 - B) 5
 - C) 7
 - D) 10
20. How can you create a generic method that accepts two arguments of different types?
 - A) public <T, U> void myMethod(T t, U u)
 - B) public void <T, U> myMethod(T t, U u)
 - C) public void myMethod<T, U>(T t, U u)
 - D) public <T extends U> void myMethod(T t, U u)

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1.	a) What is the significance of this keyword in Java? Develop a program that uses this pointer with a variable and with a function.	8	L*3	1	1
	b) Explain the difference between method overloading and method overriding?	8	L2	1	1
2.	a) Describe how the abstract and final keywords are used in Java	8	L2	1	1
	b) Demonstrate the use of access protection by creating classes with different access levels and accessing them from other packages.	8	L3	2	1
3.	a) Implement a program with multiple levels of inheritance and show how constructors are called in sequence.	8	L3	2	1
	b) What is inheritance? As a programmer, how will you identify the need for inheritance in a program? Explain how super keywords are used in inheritance.	8	L2	2	1

Unit – II

4. a)	Explain the syntax of exception handling in Java. Demonstrate with an illustration	8	L2	3	1
b)	Develop a Java program that creates a thread by implementing the Runnable interface.	8	L3	4	1
5. a)	Develop a program that handles three different types of exceptions inside one try block. Suitable catch blocks are to be written to give appropriate messages to the user.	8	L3	3	1
b)	Explain any four file handling functions of Java.	8	L2	4	1
6. a)	Develop a program to read the contents of a file, count the number of lines and words in that file. Copy the contents of the file to another file.	8	L3	4	1
b)	Demonstrate the use of throw and throws keywords with the syntax and code snippets.	8	L2	3	1

Unit – III

7. a)	Explain the difference between List and Queue interfaces in the Java Collections Framework.	8	L2	5	1
b)	Develop a Java swing program that has a text box, combo box, label, radio button and a push button. Write an action listener to the push button to display the data related to other input elements.	8	L3	5	1
8. a)	Implement a program that uses a LinkedList to store and display a list of strings.	8	L3	5	1
b)	Explain the syntax and usage of any four swing objects that are used for creating graphical user interface in Java.	8	L2	5	1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
